

GPT4RUL Reproduction and Improvement: Technical Report on LLM-based RUL Prediction

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July 2026

Abstract

This report presents an independent reproduction and systematic improvement of GPT4RUL (Tan et al., QR2MSE 2025) on the NASA C-MAPSS benchmark (FD001–FD004). The pipeline combines KMeans-based normalization, correlation–monotonicity feature selection, Patch-Channel Fusion (PCF) blocks, and a frozen GPT-2 backbone, extended with condition embedding and soft prompts. Under the reproduced configuration, GPT4RUL+CE achieves test RMSE of 14.87/12.87/13.69/13.38 across four subsets; Hybrid+Soft Prompt reaches **12.55** on FD002, outperforming the reported paper baseline on that subset. Ablation studies show that validating on only the last window per engine degrades test RMSE by 5–11 cycles, and random engine-disjoint splits are essential to avoid leakage. Code and one-click reproduction scripts are available at <https://github.com/icemilk-1/GPT4RUL-Reproduction>.

1 Method

1.1 Preprocessing

On C-MAPSS, preprocessing proceeds as follows:

- **Regime-aware normalization:** KMeans on three operating settings ($K=1$ for FD001/003; $K=6$ for FD002/004), per-cluster Z-score, then global MinMax to $[0, 1]$.

- **Feature selection:** for sensor j ,

$$Cri_j = 0.5 \cdot |r_j| + 0.5 \cdot mono_j, \quad (1)$$

keeping sensors with $Cri_j > \bar{Cri}$ (13–14 sensors), plus 3-dim condition embedding (17 features total).

- **Windowing & split:** stride=1, RUL cap 125; Random Engine Split (80/20 by engine ID) to prevent cross-set leakage.

Table 1: Per-dataset hyperparameters (Tan et al., Table 2).

Param	FD001	FD002	FD003	FD004
Window	30	64	40	64
Patch stride	4	8	8	8
PCF blocks	2	2	1	2
Mixing factor	2	2	1	1
Batch size	64	128	64	128

1.2 Architecture

Figure 1 summarizes GPT4RUL. Given $\mathbf{X} \in \mathbb{R}^{B \times T \times F}$, patching ($K=8$) feeds PCF blocks:

$$\mathbf{H}' = \mathbf{H} + \text{MLP}_{\text{patch}}(\text{LN}(\mathbf{H})), \quad (2)$$

$$\mathbf{H}'' = \mathbf{H}' + \text{MLP}_{\text{channel}}(\text{LN}(\mathbf{H}')), \quad (3)$$

followed by linear mapping to 768-dim, frozen 3-layer GPT-2, flatten, and RUL head. Hybrid Prompt RUL prepends $M=4$ learnable 768-dim soft prompt tokens. Only PCF, projections, head (and soft prompts) are trainable; GPT-2 remains frozen ($\sim 38\text{M}$ params).

1.3 Training

Adam ($lr=0.005$, weight decay 0.01), MSE loss, dropout 0.2, StepLR ($\times 0.1$ every 10 epochs), early-stop patience=10. Batch sizes: 64/128/64/128 for FD001–FD004; validation uses all engine-disjoint windows (`val_mode=all`).

2 Experiments

PyTorch, seed=42; metrics:

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (\hat{y}_i - y_i)^2}, \quad (4)$$

and asymmetric C-MAPSS Score; predictions clipped to $[0, 125]$.

Table 2 reports GPT4RUL+condition embedding; FD004 is closest to the paper (+0.70 RMSE). Table 3 compares three variants: PCF-only wins on simple FD003; Hybrid+soft prompts win on multi-regime FD001/002, with **FD002 (12.55) as the only subset beating the paper baseline among tested configs**.

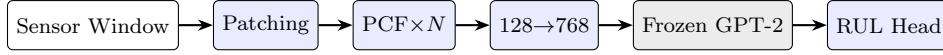


Figure 1: GPT4RUL architecture (blue: trainable; gray: frozen).

Table 2: Main reproduction vs. Tan et al. (2025).

Dataset	Paper	Reproduced	Gap
FD001	11.23	14.87	+3.64
FD002	11.05	12.87	+1.82
FD003	11.45	13.69	+2.24
FD004	12.68	13.38	+0.70

Table 3: Multi-model test RMSE comparison (lower is better).

Dataset	PCF only	GPT4RUL+CE	Hybrid+SP	Paper
FD001	14.16	14.87	13.80	11.23
FD002	13.61	12.87	12.55	11.05
FD003	12.99	13.69	13.59	11.45
FD004	14.52	13.38	13.76	12.68

3 Ablation Findings

3.1 Validation Protocol Trap

Using only the last window per engine for validation (RUL $\approx$ 0) induces a near-zero prediction shortcut, severely degrading test RMSE (Table 4).

3.2 Diagnostic Checklist

Key findings: (1) LLM value depends on task complexity; PCF-only suffices on simple FD003; (2) Soft prompts are critical to activate GPT-2; without them LLM can hurt performance; (3) *Evaluation protocol* (split and validation design) matters more than PCF micro-tuning (< 0.5 RMSE); (4) No single architecture wins on all four subsets.

Figure 4 shows how the validation protocol affects training: last-window validation locks an underfit checkpoint at epoch 1.

4 Conclusion

This report delivers an end-to-end GPT4RUL reproduction on C-MAPSS with 12+ diagnostic experiments, highlighting methodological pitfalls in LLM-based PHM. Main outcomes:

1. Reproducible open-source pipeline with one-click scripts (`reproduce.ps1`).
2. Best current result: Hybrid+Soft Prompt on FD002 (RMSE 12.55).
3. Engine-disjoint splits and full validation windows are necessary for credible evaluation.

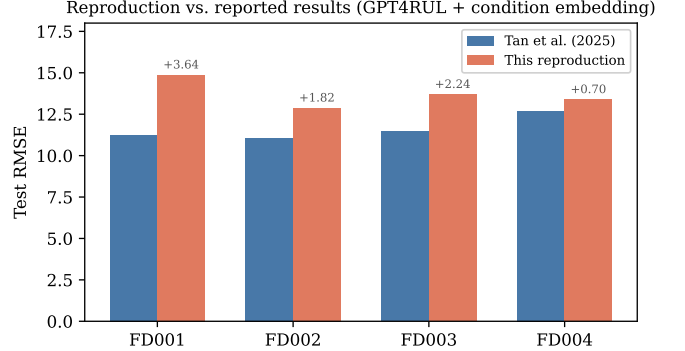


Figure 2: Reproduced vs. reported RMSE.

Table 4: Validation protocol ablation (test RMSE).

Dataset	last_window	all-val	Gain
FD001	26.19	14.87	-11.32
FD002	17.90	12.87	-5.03
FD003	23.89	13.69	-10.20

Recommendations: always split by engine ID; never validate on last-window-only subsets; encode operating conditions for multi-regime data. Future work: GPT-2 layer ablation, multi-seed reporting, and fully logged paired ablations.

Repository: <https://github.com/icemilk-1/GPT4RUL-Reproduction>

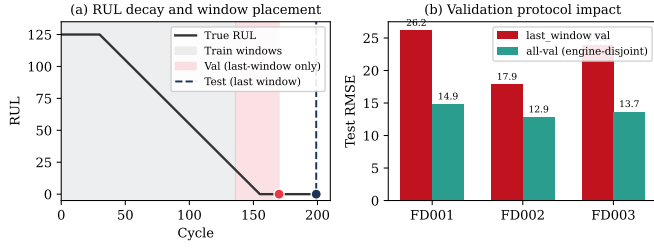


Figure 3: RUL decay and validation protocol impact.

Table 5: Selected items from 12+ reproduction diagnostics.

Item	Outcome
last_window $\rightarrow$ all val	+5-11 RMSE gain
Random Engine Split	No leakage (adopted)
Temporal split	val inflated ~ 6.6 RMSE
Condition Embedding	Improved all 4 subsets
Dual Pre-LN PCF	Aligned with MLP-Mixer
Dropout 0.3 / LR 0.002	Degraded
Grad clip off vs 1.0	Preliminary: off better
GPT-2 layer count	Pending ablation

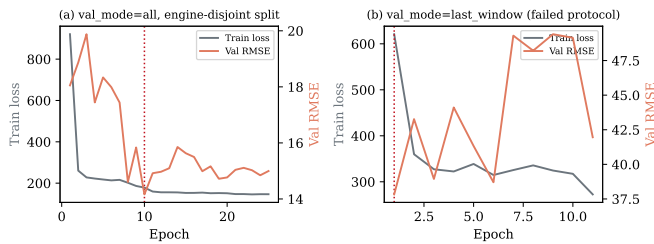


Figure 4: FD001 training dynamics: correct vs. last-window validation (early stop at epoch 1).